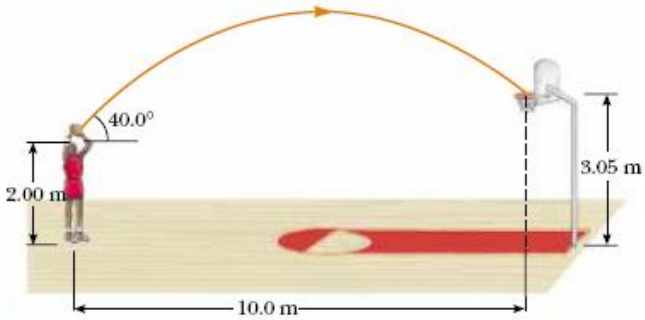
Addis Ababa Science & Technology University

Department of Physics
General Physics (Phys1001)
Worksheet on Chapter-2

(Group assignment2: #1, #9, #12, #13, #17, #21, #22, #25, #33, #34) Submission Date:_____)

1. A truck on a straight road starts from rest, accelerating at 2 m/s^2 until it reaches a speed of 20 m/s. Then the truck travels for 20 s at constant speed until the brakes are applied, stopping the truck in a uniform manner in an additional 5 s. (a) How long is the truck in motion? (b) What is the average velocity of the truck for the motion described?
2. The position of an object is given as a function of time by $x = 7t - 3t^2$, where x is in meters and t is in seconds. a) What is its instantaneous velocity at t=2s? b) Calculate its average velocity over the interval from t = 0 to t = 4 s.
3. The velocity of an object is given as a function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. a) What is its instantaneous acceleration at t=1s? b) Calculate its average acceleration over the interval from t = 0 to t = 2 s
4. An object moving with uniform acceleration has a velocity of 12 m/s in the positive x direction when its coordinate is 3 m. If its coordinate 2 s later is 5m, what is its acceleration?
5. An air plane moving horizontally at a height of 500m releases a heavy bomb to hit a target at a horizontal distance of 1000m. With what initial velocity must the plane move in order to hit the target?
6. A stone is thrown from the top of a building upward at an angle of 37^0 to the horizontal with an initial speed of 20.0 m/s. If the height of the building is 45.0 m,
   a) How long does it take the stone to reach the ground?
   b) What is the speed of the stone just before it strikes the ground?
7. a) A constant 10-N horizontal force is applied to a 20-kg cart at rest on a level floor. If friction is negligible, what is the speed of the cart when it has been pushed 8 m?
   b) If the 10-N force is applied at an angle of 45° above the horizontal. What is the speed of the cart when it has been pushed 8 m?
8. A train slows down as it rounds a sharp horizontal turn, slowing from 90.0 km/hr to 50.0 km/hr in the 15.0 s that it takes to round the bend. The radius of the curve is 150 m. Compute the acceleration at the moment the train speed reaches 50.0 km/hr. Assume it continues to slow down at this time at the same rate.

9. A basketball player who is 2.00 m tall is standing on the floor 10.0 m from the basket, as shown in the Figure below. If he shoots the ball at a 40.0° angle with the horizontal, at what initial speed must he throw so that it goes through the hoop without striking the backboard? The basket height is 3.05 m.

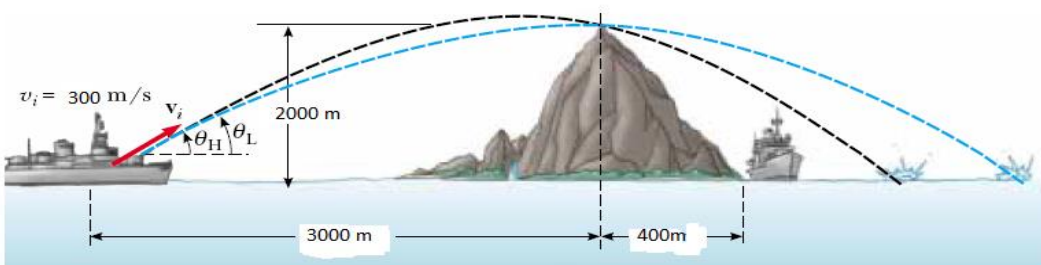


10. Quarterback throws a football straight toward a receiver with an initial speed of 20.0 m/s, at an angle of 30.0° above the horizontal. At that instant, the receiver is 20.0 m from the quarterback. In what direction and with what constant speed should the receiver run in order to catch the football at the level at which it was thrown?

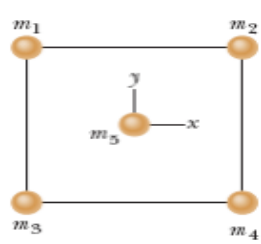
11. A car is parked on a steep incline overlooking the ocean, where the incline makes an angle of 37.0° below the horizontal. The negligent driver leaves the car in neutral, and the parking brakes are defective. Starting from rest at t = 0, the car rolls down the incline with a constant acceleration of 4.00 m/s^2, traveling 50.0 m to the edge of a vertical cliff. The cliff is 30.0 m above the ocean. Find,

(a) The speed of the car when it reaches the edge of the cliff and the time at which it arrives there,
(b) The velocity of the car when it lands in the ocean,
(c) The total time interval that the car is in motion, and
(d) The position of the car when it lands in the ocean, relative to the base of the cliff.

12. An enemy ship is on the east side of a mountain island, as shown in Fig below. The enemy ship has maneuvered to within 3000 m of the 2000 m high mountain peak and can shoot projectiles with an initial speed of 300 m/s. If the western shoreline is horizontally 400 m from the peak, what are the distances from the western shore at which a ship can be safe from the bombardment of the enemy ship?



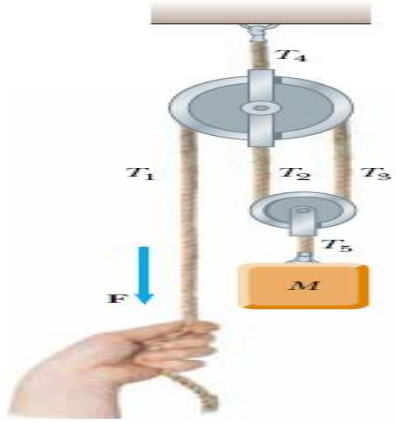
13. A square edge length 20. 0 cm is formed by four spheres of equal masses each m = 5 g**,** as shown in the Fig below. In unit-vector notation, what is the net gravitational force from them on a central sphere with mass m_5 = 2.5 g**?**



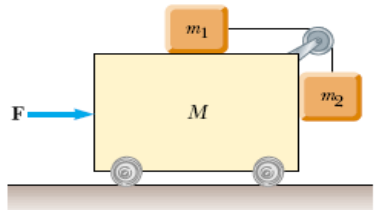
14. An object of mass M is held in place by an applied force **F** and a pulley system as shown in Figure below. The pulleys are massless and frictionless. Find:

(a) The tension in each section of rope, T_1, T_2, T_3, T_4, and T_5 and

(b) The magnitude of **F**. *Suggestion:* Draw a free-body diagram for each pulley.



15. What horizontal force must be applied to the cart shown in Fig below in order that the blocks remain stationary relative to the cart? Assume all surfaces, wheels, and pulley are frictionless. (*Hint:* Note that the force exerted by the string accelerates $m1$.)

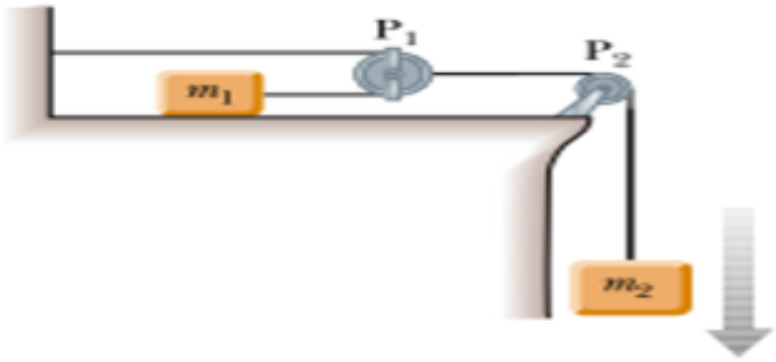


16. An object of mass M_1 on a frictionless horizontal table is connected to an object of mass m_2 through a very light pulley P_1 and a light fixed pulley P_2 as shown in Figure below.

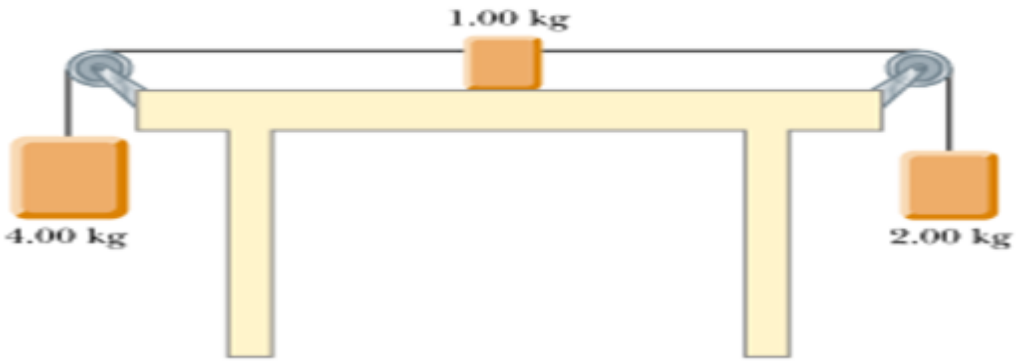
(a) If a_1 and a_2 are the accelerations of m_1 and m_2 respectively. What is the relation between these acceleration? Express

(b) The tensions in the strings and

(c) The accelerations a_1 and a_2 in terms of the masses m_1 and m_2, and g

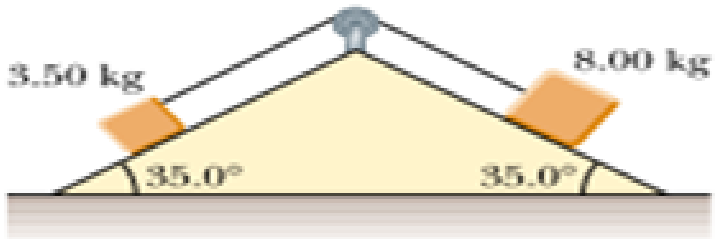


17. Three objects are connected on the table as shown in the figure below. The table is rough and has a coefficient of kinetic friction of 0.350. The objects have masses of 4.00 kg, 1.00 kg, and 2.00 kg, as shown, and the pulleys are frictionless. Draw free-body diagrams of each of the objects.
(a) Determine the acceleration of each object and their directions.
(b) Determine the tensions in the two cords.



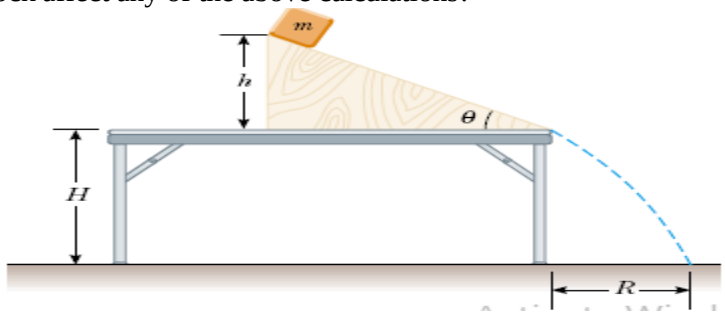
18. Three blocks are in contact with each other on a frictionless, horizontal surface, as shown in the figure below. A horizontal force **F** is applied to m_1. Take m_1 = 2.00 kg, m_2 = 3.00 kg, m_3 = 4.00 kg and **F** = 18.0 N. Draw a separate free-body diagram for each block and find
(a) The acceleration of the blocks
(b) The resultant force on each block
(c) The magnitudes of the contact forces between the blocks
(d) You are working on a construction project. A coworker is nailing up plasterboard on one side of a light partition, and you are on the opposite side, providing "backing" by leaning against the wall your back pushing on it. Every blow makes your back wood between the wall and your back. Using the situation analyzed in parts (a), (b), and (c) as a model, explains how this works to make your job more comfortable.

19. Two blocks of mass 3.50 kg and 8.00 kg are connected by a massless string that passes over a frictionless pulley as shown below. The inclines are frictionless. Find
(a) The magnitude of the acceleration of each block and
(b) The tension in the string



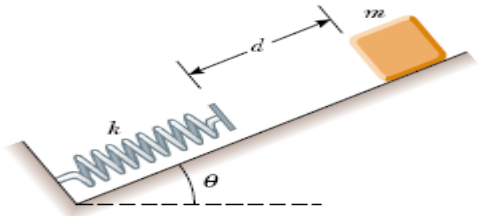
20. A block of mass m = 2.00 kg is released from rest at h = 0.500 m above the surface of a table, at the top of θ = 30 .0 ° incline as shown in Fig below. The frictionless incline is fixed on a table of height H = 2.00 m.

(a) Determine the acceleration of the block as it slides down the incline.
(b) What is the velocity of the block as it leaves the incline?
(c) How far from the table will the block hit the floor?
(d) How much time has elapsed between when the block is released and when it hits the floor?
(e) Does the mass of the block affect any of the above calculations?

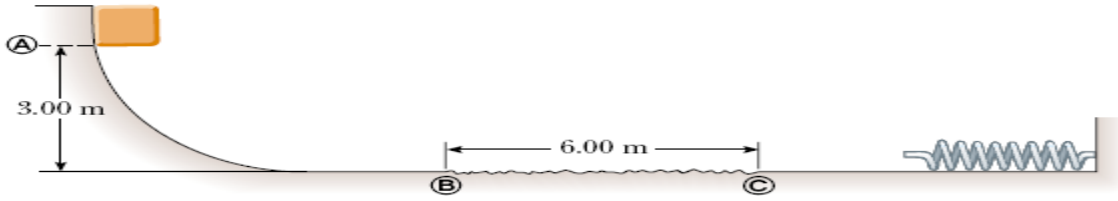


21. A bead slides without friction around a loop-the-loop (see Fig. below). The bead is released from rest at a height h = 3.50R. What is its speed at point A?

22. An object of mass m starts from rest and slides a distance d down a frictionless incline of angle θ. While sliding, it contacts an unstressed spring of negligible mass as shown in Fig below. The object slides an additional distance x as it is brought momentarily to rest by compression of the spring (of force constant k). Find the initial separation d between object and spring.

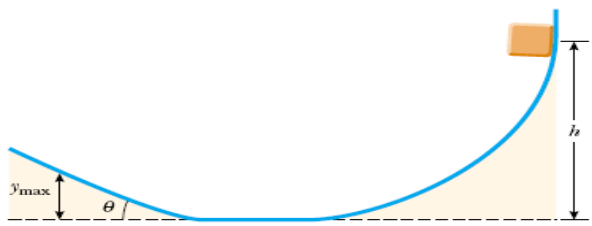


23. A 10.0-kg block is released from point A in Fig below. The track is frictionless except for the portion between points B and C, which has a length of 6.00 m. The block travels down the track, hits a spring of force constant 2250 N/m, and compresses the spring 0.30 m from its equilibrium position before coming to rest momentarily. Determine the coefficient of kinetic friction between the block and the rough surface between B and C.



24. A block slides down a curved frictionless track and then up an inclined plane as in Fig below. The coefficient of kinetic friction between block and incline is) μ_k. Use energy methods to show that the maximum height reached by the block is given as:

$$y_{max} = \frac{h}{1 + \mu_k \cot\theta}$$



25. A 1500-kg car traveling east with a speed of 25.0 m/s collides at an intersection with a 2 500-kg van traveling north at a speed of 20.0 m/s, Find the direction and magnitude of the velocity of the wreckage after the collision, assuming that the vehicles undergo a perfectly inelastic collision (that is, they stick together).

26. A Truck versus a Compact: A pickup truck with mass 1.80 x 10^3 kg traveling eastbound at +15.0 m/s, while a compact car with mass 9.00 x 10^2 kg is traveling westbound at -15.0 m/s as shown in the figure below. The vehicles collide head-on, becoming entangled.

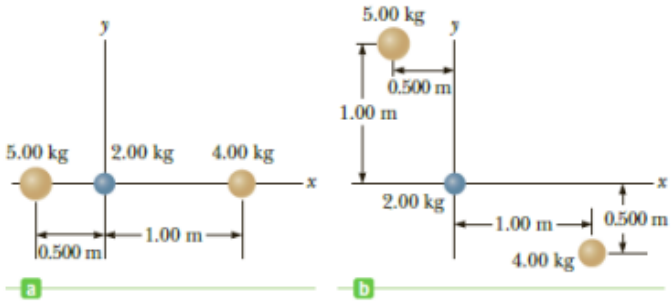
(a) Find the speed of the entangled vehicles after the collision.

(b) Find the change in the velocity of each vehicle.

(c) Find the change in the kinetic energy of the system consisting of both vehicles.

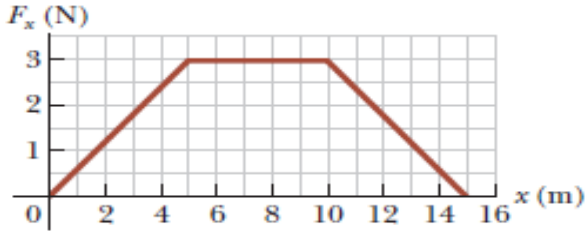
27. (a) Three objects are located in a coordinate system as shown in Fig (a). Find the center of mass.

(b) How does the answer change if the object on the left is displaced upward by 1.00 m and the object on the right is displaced downward by 0.500 m in Fig. (b)? Treat the objects as point particles.



28. Four objects are situated along the y axis as follows: a 2.00 kg object is at +3.00 m, a 3.00-kg object is at +2.50 m, a 2.50 kg object is at the origin, and a 4.00 kg object is at -0.500 m. Where is the center of mass of these objects?

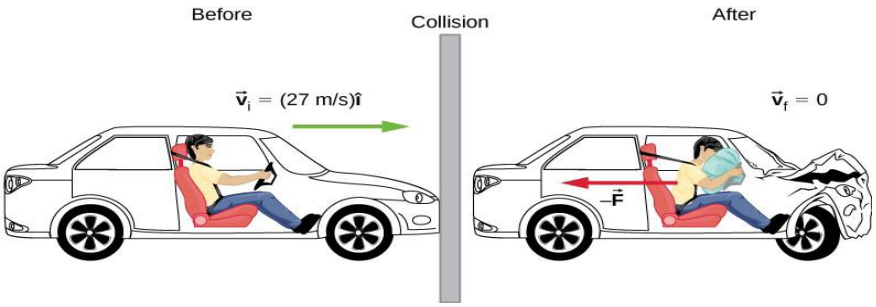
29. A particle is subject to a force F_x that varies with position as shown below. The work done by the force on the particle as it moves a) from x=0 to x=5.0 m b) from x=5.0 m to 10.0m c) from x=10 m to x=15 m d) what is the total work done by the force over the distance from x= 0 to x= 15.0 m?



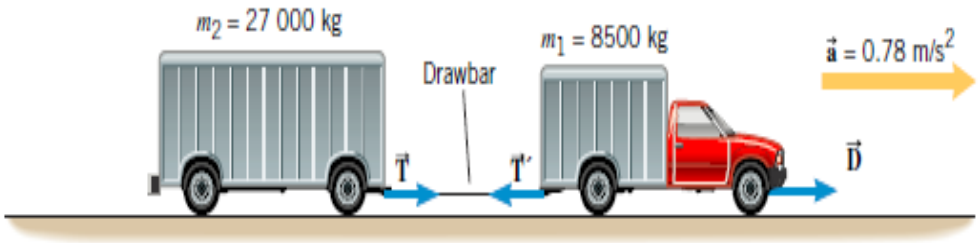
30. A car traveling at 27 m/s collides with a building. The collision with the building causes the car to come to a stop in approximately 1 second. The driver, who weighs 860 N, is protected by a combination of a variable-tension seatbelt and an airbag (Figure). (In effect, the driver collides with the seatbelt and airbag and *not* with the building.) The airbag and seatbelt slow his velocity, such that he comes to a stop in approximately 2.5 s.

a) What average force does the driver experience during the collision?

b) Without the seatbelt and airbag, his collision time would have been approximately 0.20 s. What force would he experience in this case?



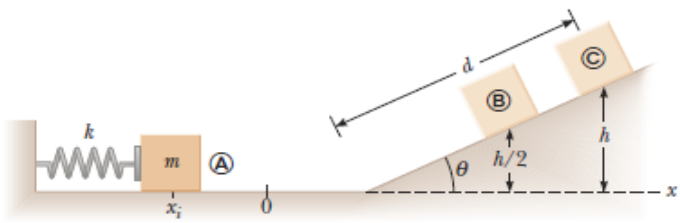
31. A truck is hauling a trailer along a level road, as figure below illustrates. The mass of the truck is m_1 = 8500 kg and that of the trailer is m_2 = 27 000 kg. The two move along the *x* axis with an acceleration of a =0.78 m/s^2. Ignoring the retarding forces of friction and air resistance and determine: **(a)** the tension **T** in the horizontal drawbar between the trailer and the truck and **(b)** the force S**D** that propels the truck forward.



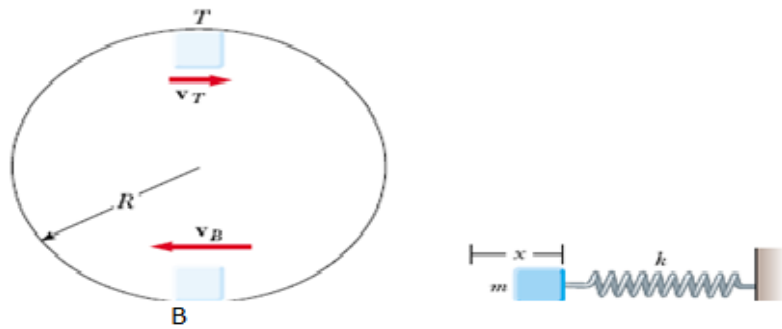
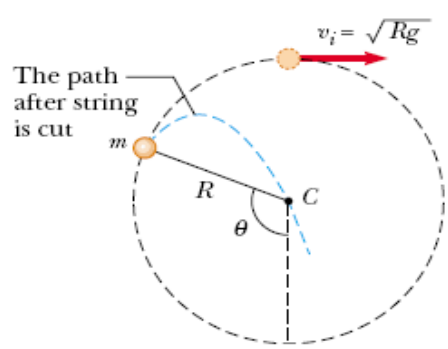
32. A 0.500-kg block rests on a horizontal, frictionless surface as in Figure below. The block is pressed back against a spring having a constant of k = 625 N/m, compressing the spring by 10.0 cm to point Ⓐ. Then the block is released.

(a) Find the maximum distance d the block travels up the frictionless incline if θ= 30°.

(b) How fast is the block going at half its maximum height?



33. A block of mass 0.500 kg is pushed against a horizontal spring of negligible mass until the spring is compressed a distance x (Fig. below). The force constant of the spring is 450 N/m. When it is released, the block travels along a frictionless, horizontal surface to point B, the bottom of a vertical circular track of radius R = 1.00 m, and continues to move up the track. The speed of the block at the bottom of the track is V_B = 12.0 m/s, and the block experiences an average friction force of 7.00 N while sliding up the track. (a) What is x? (b) What speed do you predict for the block at the top of the track? (c) Does the block actually reach the top of the track, or does it fall off before reaching the top?



34. A ball is tied to one end of a string. The other end of the string is held fixed. The ball is set moving around a vertical circle without friction, and with speed $\boldsymbol{v_i = \sqrt{Rg}}$ at the top of the circle, as in Fig given below. At what angle θ should the string be cut so that the ball will then travel through the center of the circle?